Candidate Methodological Directions for the Master's Thesis

Application: Opioid Distribution in ARCOS Data

Fafa Tchorly
Department of Mathematics and Statistics, University of North Florida
Advisor : Dr. Michelle DeDeo • 17th October 2026

I have narrowed the thesis to two methodological directions. Each starts from a classroom topic, asks a statistical question that simulation can answer where the truth is known, and uses the opioid data as the application.

Common Setting: Let y_{tjkl} denote log morphine milligram equivalents (MME) per 1,000 residents in month t, ZIP code j, county k and region l. The baseline four-level model is

$$y_{tjkl} = \mathbf{x}_{tjkl}^{\top}\boldsymbol{\beta} + v_l + u_{kl} + w_{jkl} + \varepsilon_{tjkl}, \tag{1}$$

with independent $v_l \sim N(0, \sigma_R^2)$, $u_{kl} \sim N(0, \sigma_C^2)$, $w_{jkl} \sim N(0, \sigma_Z^2)$ and $\varepsilon_{tjkl} \sim N(0, \sigma_\varepsilon^2)$, where $\mathbf{x}_{tjkl}$ may contain covariates measured at any level (for example income).

Direction 1: BIC for Multilevel Models with Three or Four Levels

Classroom starting point: Linear mixed models (ML/REML) and model selection with AIC and BIC.

Statistical question: The Bayesian information criterion $\text{BIC} = -2\,\ell(\hat{\theta}) + k \log n$ comes from a Laplace approximation to the marginal likelihood,

$$\log p(\mathbf{y} \mid M) = \ell(\hat{\theta}) - \tfrac{1}{2} \log\det I(\hat{\theta}) + O_p(1), \tag{2}$$

where I is the Fisher information. Replacing $\log\det I$ by $k \log n$ is justified only if the information about every parameter grows at the same rate n. In model (1) it does not: information about a parameter grows with the number of units at its own level, and software packages disagree on whether n should be the number of observations or of clusters. What is the correct penalty for three- and four-level nested models, and does it select the right levels when the top level has few units?

Related Work: Jones (2011) replaced n by an effective sample size for correlated data; Delattre et al. (2014) used the number of clusters for random effects and the total sample size for fixed effects; Lorah and Womack (2019) compared choices of n by simulation for two-level models and conjectured, without derivation, a level-specific penalty for three levels; Cho et al. (2024) derived the penalty for level-specific selection in a two-level nested model and leave higher-level designs for future study; and Greven and Kneib (2010) showed that marginal AIC favours dropping random effects.

Preliminary Result: Consider the balanced three-level case (months within Z ZIP codes within each of C counties, T months, $N = CZT$). Each county's covariance matrix is $V = \sigma_\varepsilon^2 I_{ZT} + \sigma_Z^2 (I_Z \otimes J_T) + \sigma_C^2 J_{ZT}$, with J a matrix of ones, and has only three distinct eigenvalues,

$$\lambda_1 = \sigma_\varepsilon^2, \qquad \lambda_2 = \sigma_\varepsilon^2 + T\sigma_Z^2, \qquad \lambda_3 = \sigma_\varepsilon^2 + T\sigma_Z^2 + ZT\sigma_C^2, \tag{3}$$

with total multiplicities $CZ(T-1)$, $C(Z-1)$ and C. Working through the information matrix (variance components positive, each covariate varying at a single level, and $C, Z, T \to \infty$) gives

$$\mathrm{BIC}_3 = -2\,\ell(\hat{\theta}) + (p_1+1)\log N + (p_2+1)\log(CZ) + (p_3+1)\log C, \tag{4}$$

where p_1, p_2, p_3 count the month-, ZIP- and county-level fixed effects (intercept at the county level) and each "+1" is the variance component of that level.It implies that removing a level also changes the penalties on the remaining parameters, which existing criteria ignore.

Competing Methods : BIC with total N; BIC with number of clusters; Jones's effective sample size; marginal and conditional AIC; boundary-corrected and exact restricted likelihood ratio tests.

Simulation: Data generated from (1) with chosen variance components set to zero, varying the number of top-level units (e.g. 4, 8, 15), units per level, intraclass correlations and balance. Outcomes: proportion of correctly selected structures, false-drop and false-keep rates per level, and coverage of fixed-effect intervals after selection.

Opioid Application. Which of the ZIP, county and region levels the data support, the share of variation at each level, and whether conclusions about income and rurality depend on the levels kept; sensitivity to AR(1) errors and to ZIP codes that cross county lines.

Key Articles: Jones (2011); Delattre et al. (2014); Greven and Kneib (2010); Lorah and Womack (2019); Cho et al. (2024).

Direction 2: Multilevel Random Change-Point Models

Classroom Starting Point: Regression with a structural break, $y_t = \mathbf{x}_t^\top \boldsymbol{\beta} + \delta\,\mathbf{1}\{t \geq \tau\} + \varepsilon_t$, tested with the Chow F-statistic $F_T(\tau)$ at a known date τ. When τ is unknown, Andrews (1993) uses

$$\sup F_T = \sup_{\tau \in [\pi T,\,(1-\pi)T]} F_T(\tau), \tag{5}$$

whose limiting null distribution is non-standard.

Statistical Question: The break date need not be common to all units: ZIP codes may respond to a policy change at different times, and the timing may cluster within counties and regions.I propose to model break dates as nested random effects,

$$y_{tjkl} = \mathbf{x}_{tjkl}^\top \boldsymbol{\beta} + w_{jkl} + \delta_{jkl}\,\mathbf{1}\{t \geq \tau_{jkl}\} + \varepsilon_{tjkl}, \tag{6}$$

$$\tau_{jkl} = \tau_0 + a_l + b_{kl} + c_{jkl}, \qquad a_l \sim N(0, \sigma^2_{\tau,R}), \quad b_{kl} \sim N(0, \sigma^2_{\tau,C}), \quad c_{jkl} \sim N(0, \sigma^2_{\tau,Z}), \tag{7}$$

with τ_{jkl} discretised to the monthly grid and the jump sizes δ_{jkl} allowed to vary by level. Can τ_0 and the timing variances $\sigma^2_{\tau,R}$, $\sigma^2_{\tau,C}$, $\sigma^2_{\tau,Z}$ be estimated jointly, and under what conditions on the series length and numbers of units are they identifiable?

Related Work: Bai and Perron (1998) extended estimation and testing to multiple breaks; Bai (2010) estimated a common break date in panel data; Okui and Wang (2021) allowed breaks to differ across latent groups using grouped fixed effects and adaptive group fused lasso; Lumsdaine et al. (2023) allowed group membership itself to change at a break; and Brilleman et al. (2017) fitted a Bayesian piecewise linear mixed model with a random change point at the individual level.

Theory to Develop: The likelihood of (6)–(7) is non-smooth in the discrete break dates, so standard asymptotics do not apply. Estimation would be hierarchical Bayesian, marginalising over the discrete τ_{jkl} within MCMC, together with conditions for identifiability with short monthly series and the effects of autocorrelation and heteroscedasticity.

Competing Methods: Bai–Perron estimation for each ZIP code separately; the common-break panel estimator of Bai (2010); grouped heterogeneous breaks (Okui and Wang, 2021); a single-level random change point (Brilleman et al., 2017).

Simulation: Break dates generated from (7), varying the numbers of ZIP codes, counties and regions, series length, jump size, timing variances, and the presence of autocorrelation and heteroscedasticity. Outcomes: bias and RMSE of the break dates, coverage of credible intervals, recovery of the timing variances, and comparison with the competing methods.

Opioid Application: Hydrocodone combination products moved from Schedule III to Schedule II on 6 October 2014 (Drug Enforcement Administration, 2014). The model estimates when each ZIP code's hydrocodone supply shifted, how much the timing varies across ZIP codes, counties and regions, and whether rurality or income explains the timing.

Key Articles: Bai and Perron (1998); Bai (2010); Okui and Wang (2021); Lumsdaine et al. (2023); Brilleman et al. (2017).

Comparison and Preferred Plan

	Direction 1: Multilevel BIC	**Direction 2: Nested Change Points**
Main contribution	New derivation (theory)	New model and estimation (Computation)
Evidence the question is open	Strong: Stated as future work by Cho et al. (2024)	Moderate: No nested version found so far
Simulation cost	Fast (closed-form likelihood)	Slower (MCMC for every simulated data set)
Main risk	Derivation may not close for unbalanced designs	Identifiability with short series
Fallback	Balanced case plus simulation of conjectured penalties	Two-level model, or known break date

(i) **Preferred(Direction 1)** Derive the BIC penalty for three- or four-level models, starting from (4), and evaluate it by simulation against existing criteria.

(ii) **Alternative (Direction 2)** With the known date $D_t = \mathbf{1}\{t \geq \text{Given study period}\}$, extend (1) with a nested random jump,

$$y_{tjkl} = \mathbf{x}_{tjkl}^{\top}\boldsymbol{\beta} + v_l + u_{kl} + w_{jkl} + \big(\delta_0 + \delta_l + \delta_{kl} + \delta_{jkl}\big)D_t + \varepsilon_{tjkl}. \tag{8}$$

This is still a linear mixed model.

References

[1] D. W. K. Andrews. Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4):821–856, 1993.

[2] J. Bai. Common breaks in means and variances for panel data. *Journal of Econometrics*, 157(1):78–92, 2010.

[3] J. Bai and P. Perron. Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78, 1998.

[4] S. L. Brilleman, L. D. Howe, R. Wolfe, and K. Tilling. Bayesian piecewise linear mixed models with a random change point: An application to BMI rebound in childhood. *Epidemiology*, 28(6):827–833, 2017. doi: 10.1097/EDE.0000000000000723.

[5] S.-J. Cho, H. Wu, and M. Naveiras. The effective sample size in Bayesian information criterion for level-specific fixed and random-effect selection in a two-level nested model. *British Journal of Mathematical and Statistical Psychology*, 77:289–315, 2024. doi: 10.1111/bmsp.12327.

[6] M. Delattre, M. Lavielle, and M.-A. Poursat. A note on BIC in mixed-effects models. *Electronic Journal of Statistics*, 8(1):456–475, 2014. doi: 10.1214/14-EJS890.

[7] Drug Enforcement Administration. Schedules of controlled substances: Rescheduling of hydrocodone combination products from Schedule III to Schedule II. *Federal Register*, 79(163):49661, 2014.

[8] S. Greven and T. Kneib. On the behaviour of marginal and conditional AIC in linear mixed models. *Biometrika*, 97(4):773–789, 2010. doi: 10.1093/biomet/asq042.

[9] R. H. Jones. Bayesian information criterion for longitudinal and clustered data. *Statistics in Medicine*, 30:3050–3056, 2011. doi: 10.1002/sim.4323.

[10] J. Lorah and A. Womack. Value of sample size for computation of the Bayesian information criterion (BIC) in multilevel modeling. *Behavior Research Methods*, 51:440–450, 2019. doi: 10.3758/s13428-018-1188-3.

[11] R. L. Lumsdaine, R. Okui, and W. Wang. Estimation of panel group structure models with structural breaks in group memberships and coefficients. *Journal of Econometrics*, 233(1):45–65, 2023.

[12] R. Okui and W. Wang. Heterogeneous structural breaks in panel data models. *Journal of Econometrics*, 220(2):447–473, 2021. doi: 10.1016/j.jeconom.2020.04.009.